# MATHS MANTHAN

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**Subject:** Mathematics | **CBSE CLASS 10 – TEST (SET – B)** | **Time:** 45 Minutes

**Class:** Class X (CBSE) | **Chapter 11: Areas Related to Circles** | **MM: 20**

**Document Type:** Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

**CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:**

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

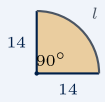
Official Evaluation Key | Maths Manthan Meerut

## SECTION A: MULTIPLE CHOICE QUESTIONS MATHS MANTHAN | (5 × 1 = 5 Marks)

**Q1.** What is the length of the arc of a sector of a circle with radius 14 cm and central angle 90°? [1]

(A) 22 cm (B) 44 cm (C) 88 cm (D) 11 cm

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]




**Step 1: Arc Length Formula** [0.5 Mark]

Given radius $r = 14$ cm, central angle $\theta = 90°$.

$$l = \frac{\theta}{360°} \times 2\pi r$$

**Step 2: Substitution and Evaluation** [0.5 Mark]

$$l = \frac{90°}{360°} \times 2 \times \frac{22}{7} \times 14 = \frac{1}{4} \times 2 \times 22 \times 2 = 22 \text{ cm}$$

**Correct Option:** (A) 22 cm

Length of the arc = **22** cm

**Q2.** A sector of a circle with central angle 120° and area $\frac{264}{7}$ cm$^2$ is cut from a circle. The radius of the circle is: [1]

(A) 6 cm (B) 5 cm (C) 7 cm (D) 12 cm

**▶ STEP-BY-STEP SOLUTION:** [1 Mark]




**Step 1: Sector Area Formula** [0.5 Mark]

$$\text{Area} = \frac{\theta}{360^\circ} \times \pi r^2 \implies \frac{120^\circ}{360^\circ} \times \frac{22}{7} \times r^2 = \frac{264}{7}$$

**Step 2: Solving for $r$** [0.5 Mark]

$$\frac{1}{3} \times \frac{22}{7} \times r^2 = \frac{264}{7} \implies \frac{22}{21} r^2 = \frac{264}{7}$$

$$r^2 = \frac{264}{7} \times \frac{21}{22} = \frac{264}{22} \times 3 = 12 \times 3 = 36 \implies r = 6 \text{ cm}$$

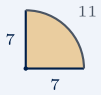
**Correct Option: (A) 6 cm**

Radius of the circle $r = \mathbf{6}$ cm

**Q3.** If the sector of a circle with diameter 14 cm makes an angle 90° at the centre, then the perimeter of the sector is: [1]

(A) 25 cm (B) 11 cm (C) 36 cm (D) 22 cm

**▶ STEP-BY-STEP SOLUTION:** [1 Mark]




**Step 1: Radius and Arc Length** [0.5 Mark]

Diameter $d = 14$ cm $\implies r = 7$ cm. Central angle $\theta = 90^\circ$.

$$l = \frac{90^\circ}{360^\circ} \times 2\pi r = \frac{1}{4} \times 2 \times \frac{22}{7} \times 7 = 11 \text{ cm}$$

**Step 2: Total Perimeter** [0.5 Mark]

$$\text{Perimeter} = l + 2r = 11 + 2(7) = 11 + 14 = 25 \text{ cm}$$

**Correct Option: (A) 25 cm**

Perimeter of sector = **25** cm

**Q4.** If the area of a sector is one-twelfth that of a complete circle, then the central angle of the sector is: [1]

(A) 36° (B) 30° (C) 60° (D) 45°

**▶ STEP-BY-STEP SOLUTION:** [1 Mark]

**Step 1: Ratio of Sector to Circle Area** [0.5 Mark]

$$\frac{\text{Area of sector}}{\text{Area of circle}} = \frac{\theta}{360^\circ} = \frac{1}{12}$$

**Step 2: Solving for $\theta$** [0.5 Mark]

$$\theta = \frac{360^\circ}{12} = 30^\circ$$

**Correct Option: (B) 30°**

Central angle of the sector $\theta = \mathbf{30}^\circ$

**Q5.** If an arc subtends an angle of $90^\circ$ at the centre of a circle, then the ratio of its length to the circumference of the circle is: [1]

(A) 2 : 3 (B) 1 : 4 (C) 4 : 1 (D) 1 : 3

► **STEP-BY-STEP SOLUTION:** [1 Mark]

**Step 1: Arc Length to Circumference Ratio** [0.5 Mark]

$$\frac{\text{Arc length } l}{\text{Circumference } C} = \frac{\frac{\theta}{360^\circ} \times 2\pi r}{2\pi r} = \frac{\theta}{360^\circ}$$

**Step 2: Substitution for $\theta = 90^\circ$** [0.5 Mark]

$$\frac{l}{C} = \frac{90^\circ}{360^\circ} = \frac{1}{4} = 1 : 4$$

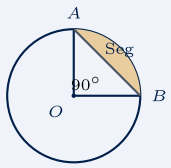
**Correct Option:** (B) 1 : 4

Ratio of arc length to circumference = **1 : 4**

## SECTION B: Short Answer Questions MATHS MANTHAN | (3 × 2 = 6 Marks)

**Q6.** A chord of a circle of radius 14 cm subtends an angle of $90^\circ$ at the centre of the circle. Find the area of the corresponding minor segment of the circle. [Use $\pi = \frac{22}{7}$] [2]

► **STEP-BY-STEP SOLUTION:** [2 Marks]




**Step 1: Area of Quadrant $OAB$** [1 Mark]

Radius $r = 14$ cm, central angle $\theta = 90^\circ$.

$$\text{Area}_{\text{quadrant}} = \frac{1}{4}\pi r^2 = \frac{1}{4} \times \frac{22}{7} \times 14 \times 14 = \frac{1}{4} \times 616 = 154 \text{ cm}^2$$

**Step 2: Area of $\triangle OAB$ and Minor Segment** [1 Mark]

Since $\angle AOB = 90^\circ$:

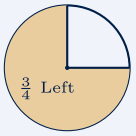
$$\text{Area}(\triangle OAB) = \frac{1}{2} \times 14 \times 14 = 98 \text{ cm}^2$$

$$\text{Area}_{\text{segment}} = \text{Area}_{\text{quadrant}} - \text{Area}(\triangle OAB) = 154 - 98 = 56 \text{ cm}^2$$

Area of the minor segment = **56 cm²**

**Q7.** From a circular sheet of paper of radius 70 cm, a quadrant is cut out. Find the area of the remaining sheet. $\left[\text{Use } \pi = \frac{22}{7}\right]$ [2]

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]




**Step 1: Total Area of Circular Sheet** [1 Mark]
Radius $r = 70$ cm.

$$\text{Total Area} = \pi r^2 = \frac{22}{7} \times 70 \times 70 = 22 \times 10 \times 70 = 15400 \text{ cm}^2$$

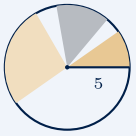
**Step 2: Area of Remaining Sheet** [1 Mark]
When 1 quadrant ($\frac{1}{4}$ of the circle) is removed, $\frac{3}{4}$ of the circle remains:

$$\text{Remaining Area} = \frac{3}{4} \times \text{Total Area} = \frac{3}{4} \times 15400 = 3 \times 3850 = 11550 \text{ cm}^2$$

Area of the remaining sheet = **11550 cm²**

**Q8.** Three sectors of a circle of radius 5 cm, having central angles 35°, 50°, and 95° at the centre, are shaded. Find the total area of the shaded region. $\left[\text{Use } \pi = \frac{22}{7}\right]$ [2]

▶ **STEP-BY-STEP SOLUTION:** [2 Marks]




**Step 1: Total Angle Swept by Shaded Sectors** [1 Mark]

$$\theta_{\text{total}} = 35^\circ + 50^\circ + 95^\circ = 180^\circ$$

Since $\theta_{\text{total}} = 180^\circ$, the combined sectors form exactly a semicircle. **Step 2: Area Calculation** [1 Mark]
Radius $r = 5$ cm.

$$\text{Total Shaded Area} = \frac{180^\circ}{360^\circ} \times \pi r^2 = \frac{1}{2} \times \frac{22}{7} \times 5^2 = \frac{11 \times 25}{7} = \frac{275}{7} \approx 39.29 \text{ cm}^2$$

Total area of the shaded region = $\frac{275}{7}$ cm² ≈ **39.29 cm²**

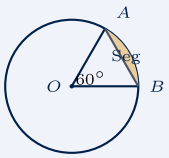
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## SECTION C: Long Answer Questions

MATHS MANTHAN | (3 × 3 = 9 Marks)

**Q9.** A chord of a circle of radius 14 cm subtends an angle of 60° at the centre. Find the area of the corresponding minor segment of the circle. [Use $\pi = 3.14$ and $\sqrt{3} = 1.73$] [3]

► **STEP-BY-STEP SOLUTION:** [3 Marks]




**Step 1: Area of Sector $OAB$** [1 Mark]
Radius $r = 14$ cm, central angle $\theta = 60°$.

$$\text{Area}_{\text{sector}} = \frac{60°}{360°} \times \pi r^2 = \frac{1}{6} \times 3.14 \times 14^2 = \frac{1}{6} \times 3.14 \times 196 = \frac{615.44}{6} \approx 102.57 \text{ cm}^2$$

**Step 2: Area of Equilateral $\triangle OAB$** [1 Mark]
Since $OA = OB = 14$ cm and $\angle AOB = 60°$, $\triangle OAB$ is equilateral.

$$\text{Area}(\triangle OAB) = \frac{\sqrt{3}}{4} r^2 = \frac{1.73}{4} \times 196 = 1.73 \times 49 = 84.77 \text{ cm}^2$$

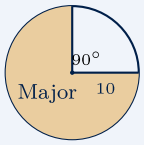
**Step 3: Area of Minor Segment** [1 Mark]

$$\text{Area}_{\text{segment}} = \text{Area}_{\text{sector}} - \text{Area}(\triangle OAB) = 102.57 - 84.77 = 17.80 \text{ cm}^2$$

Area of the minor segment = **17.80 cm²**

**Q10.** An arc of a circle of radius 10 cm subtends a right angle (90°) at the centre of the circle. Find the area of the corresponding major sector. [Use $\pi = 3.14$] **[3]**

► **STEP-BY-STEP SOLUTION:** [3 Marks]




**Step 1: Central Angle for Major Sector** [1 Mark]
Minor sector angle $\theta = 90°$.

$$\text{Major sector angle } \theta' = 360° - 90° = 270°$$

**Step 2: Area of Major Sector Formula** [1 Mark]
Radius $r = 10$ cm.

$$\text{Area}_{\text{major}} = \frac{\theta'}{360°} \times \pi r^2 = \frac{270°}{360°} \times 3.14 \times 10^2 = \frac{3}{4} \times 314$$

**Step 3: Calculation** [1 Mark]

$$\text{Area}_{\text{major}} = 3 \times 78.5 = 235.5 \text{ cm}^2$$

*(Alternatively: Total Area − Minor Area* $= 314 - \frac{1}{4}(314) = 314 - 78.5 = 235.5$ *cm².)*

Area of the major sector = **235.5 cm²**

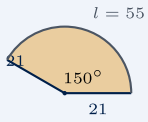
**Q11.** A sector is cut from a circle of radius 21 cm. The central angle of the sector is 150°. Find:

(i) the length of the arc of this sector, and

(ii) the area of the sector. $\left[\text{Use } \pi = \frac{22}{7}\right]$

[3]

▶ **STEP-BY-STEP SOLUTION:** [3 Marks]




**(i) Length of the Arc of the Sector** [1.5 Marks]

Radius $r = 21$ cm, central angle $\theta = 150°$.

$$l = \frac{\theta}{360°} \times 2\pi r = \frac{150°}{360°} \times 2 \times \frac{22}{7} \times 21 = \frac{5}{12} \times 2 \times 22 \times 3$$

$$= \frac{5}{12} \times 132 = 5 \times 11 = 55 \text{ cm}$$

**(ii) Area of the Sector** [1.5 Marks]

$$\text{Area} = \frac{\theta}{360°} \times \pi r^2 = \frac{150°}{360°} \times \frac{22}{7} \times 21 \times 21 = \frac{5}{12} \times 22 \times 3 \times 21$$

$$= \frac{5}{12} \times 1386 = \frac{5 \times 231}{2} = \frac{1155}{2} = 577.5 \text{ cm}^2$$

*(Alternatively: Area* $= \frac{1}{2}lr = \frac{1}{2} \times 55 \times 21 = 577.5\ cm^2$*.)*

**(i) Length of Arc = 55 cm; (ii) Area of Sector = 577.5 $\text{cm}^2$**

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